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State Space Modeling of Drilling Data to Automatically Infer Lithology

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Abstract

Surface drilling data contains a wealth of information related to rock petrophysical and mechanical properties that can be harnessed for near real-time reservoir characterization. This work provides a look-ahead for the bit to anticipate changes in lithology rapidly and with great fidelity. The forecasting discussed in this paper is based on state-space modeling that is probabilistic and provides a distribution over possible lithologies rather than a deterministic outcome. This allows for the optimization of drilling operations at the surface to simultaneously maximize rate of penetration (ROP) and minimize bit wear.

State-space models are powerful in sequentially modeling dynamic systems that are partially observable and evolve over time. In our study, the states represent the underlying petrophysical properties (or lithologies) which are rarely observed in real-time. The observations are represented by real-time surface drilling data. Our framework predicts the future sequence of lithologies, taking into account both the past sequence of observations and the inferred transition dynamics between lithologies.

We evaluate our state space model on two field-case studies: The first is the well-known Volve Field data provided by Equinor and the second is an unconventional play in Oklahoma. In both case studies, we demonstrate how the state space models can use surface drilling data to infer changes in lithology in the subsurface. From these transitions in lithology, we can derive a probabilistic model to forecast the likely sequence of states (or lithologies) the bit is likely to see soon. Remarkably, we achieve high accuracies in the near-term; however, these accuracies diminish for longer-range predictions. Nevertheless, for the drilling crew, the information we provide by looking ahead of the bit is valuable information to anticipate and prepare for changes to drilling strategy, mud weight, weight-on-bit and other operational variables.

State-space models are a relatively underexplored tool for modeling drilling systems that evolve dynamically over time. While autonomous drillers have reduced the burden on drilling crews, anticipating lithology changes can provide valuable information for geosteering applications, to optimize rate-of-penetration and to prevent excessive bit wear.

Introduction

Formations are targeted for rock properties that are favorable for production or stimulation. It is of vital importance that the geosteering team ensure the bit stays within the target formation. Target formations are often identified from seismic data or offset wells and mapped in 3D space (Solomon et al. 1994). However,

the resolution and uncertainties associated with these data types makes geosteering a challenging task. Geosteering teams are generally equipped with measurements-while-drilling (MWD) sensors that measure formation properties during the drilling process to help guide decisions about steering to stay within zone. Often, gamma ray measurements are utilized to identify the target formation, and deviations away from expected gamma ray trends can indicate drilling out of zone (Mottahedeh 2008). While these systems do provide vital information to the geosteering team, there are challenges associated with the use of MWD data. The MWD sub is generally placed farther back, sometimes as much as 30-90 feet, behind the bit (Moore et al. 1998). Fig. 1 shows a normal setup of an MWD sub showing the distances between the bit and the MWD sub.

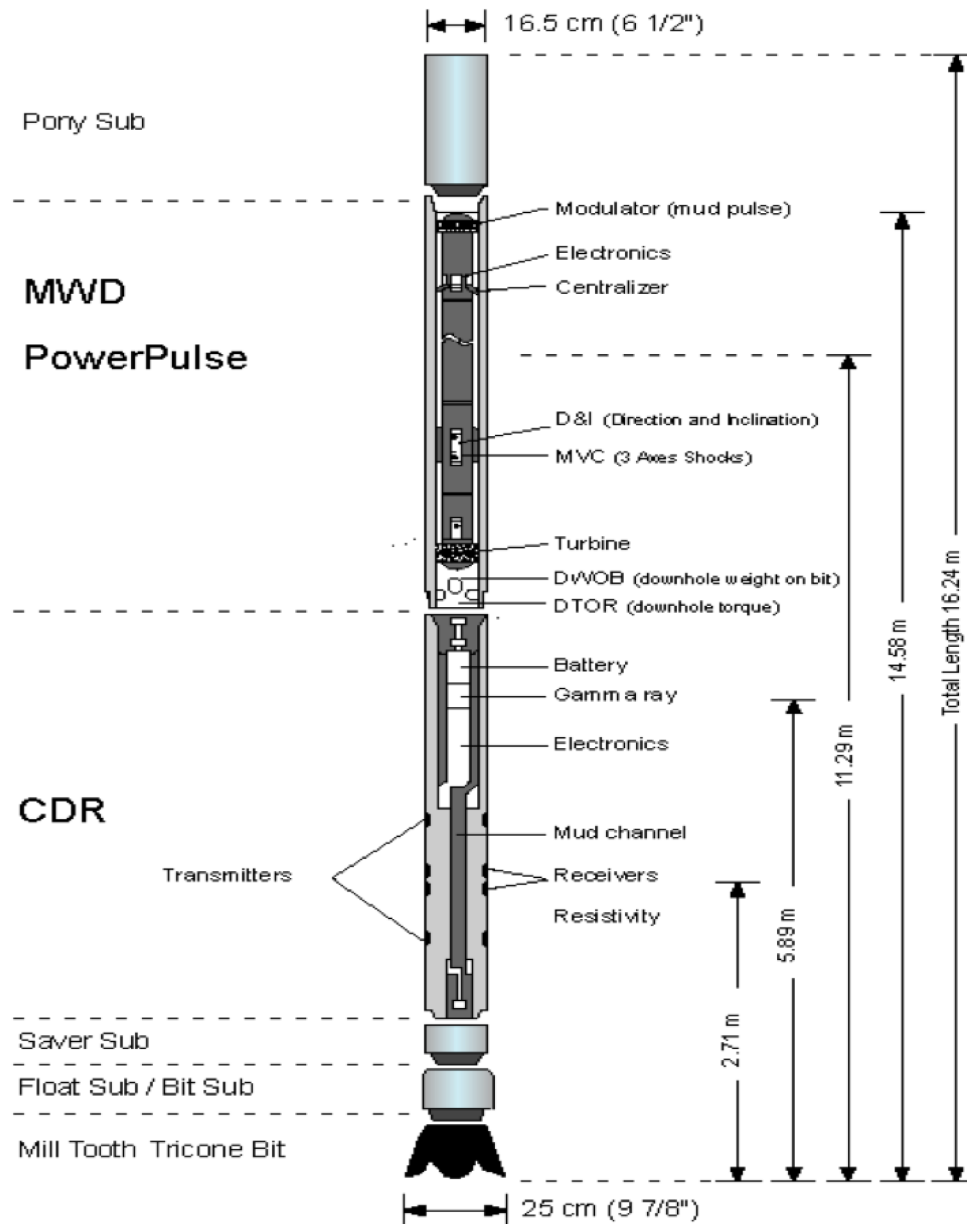


Figure 1—A figure of a typical MWD set up with various logging tools for measuring formation properties to inform drilling decisions. As can be seen from the figure, the sensors are often placed at different distances from the bit. As a result, there will be a lag from the data received from the logging tools, and the position of the bit. (Moore et al. 1998)

This distance implies that there can be a large time- and depth-lag between the MWD measurements and the location of the bit. Additionally, the mud pulse telemetry associated with MWD can take additional time

for the information to reach the surface recorders. Consequently, if the bit is out of the target zone, there may be a significant time delay before this is relayed to the surface. Information from a MWD sub can also be relayed to the surface by wired drill pipe that can transfer data quickly with high fidelity (Wilson 2013). Wired drill pipe provides a solution to the time-delay issue of MWD but can be expensive and is still depth-delayed with respect to bit performance.

Mud logging is another process used to identify formation properties during the drilling process. Initially, mud logging was a process of extracting gasses from the mud returning from downhole to identify zones with combustible hydrocarbon gasses (Whittaker 1987). Identifying hydrocarbon gasses within the return mud can signal productive formations with producible hydrocarbons. Rock cuttings from the bit are brought to surface in the return mud as well. These cuttings can be analyzed by a geologist to create a lithology log and can provide rapid analyses of reservoir properties such as porosity and permeability during drilling (Whittaker 1987). Information from mud logging can also be used to inform the geosteering team about the position of the bit within the various lithologic units, as well as identifying zones of various reservoir quality to improve landing zone determination. However, while mud logging can provide important information about formation properties, there is a lag associated with data collection because of the delay in cuttings retrieval. Additionally, in horizontal laterals, it is possible that the retrieval of cuttings may be compromised.

Due to the challenges with the time- and depth-lag and frequency of data recording with MWD and mud logging data, there is a real opportunity to explore the use of real-time surface drilling data to detect lithologies at the bit which is the focus of the work in this paper. These methods have the potential to provide the geosteering team with higher frequency data in real-time to accelerate the decision-making process.

Inoue et al. (2019) presents a method of identifying lithologies from MWD data using artificial neural networks (ANN). Lithology is determined after drilling using surface torque data to identify three lithologies: sand, silt-clay, and volcanic ash. Statistical moments (mean, standard deviation, skewness, kurtosis) and frequency content of the surface torque are used as features for model training. These features are correlated to lithology to help the model learn to classify the various rocktypes. In their work, total classification accuracy achieved was 96% with a low number of false positives. However, when the rocktypes are more similar, such high prediction accuracies may not be possible. Inoue et al. (2019) also presents an application in another scenario with seven relatively similar lithologies where the classification accuracy is observed to be much lower, and the number of false positives is relatively higher.

Romanenkova et al. (2019) use a change point detection algorithm to identify sand and shale sequences, and the transitions from one rocktype to another. They use a gradient boosted decision tree as a classifier to identify rocktypes from well log data. A change point detection algorithm is applied to drilling data such as rate-of-penetration (ROP), weight-on-bit (WOB) and torque. and the occurrence of change points is compared to transitions between the rocktypes seen on the well log. Thin layers within the rocktype sequence along the lateral are "dropped" or replaced with the previous majority rocktype with the assumption that the layers would be too thin for the change point detection algorithms to detect. The thickness of the thin beds is used as a tuning parameter for the change point detection algorithm and is selected as layers 1.5m or smaller. They use three change point detection algorithms: cumulative sums (Page 1954), Shiryaev-Roberts statistics (Roberts 1966), and posterior probabilities statistics (Smith 1975). The accuracy of the algorithm is measured by the number of predicted rocktype transitions divided by the number of true transitions. The change point detection method for lithology identification is shown to be quite effective, especially with the removal of thin layers.

The geosteering team is required to make decisions to adjust the position of the bit to stay within the target formation. Such decisions involve steering the bit up or down within the formation or making no change and following the current bit trajectory. These decisions are made in real-time and can be prone to error. Poor bit trajectory monitoring can lead the bit out of the target zone leading to contact with unproductive sections of the reservoir. Kullawan et al. (2017) describes an approach using discretized stochastic dynamic programming (DSDP) that allow the driller to keep a greater percentage of the well within the target zone.

For this approach, decision stages are predefined along the planned wellbore trajectory. These decision stages are influenced using estimates of the bed boundary depths and locations of faults and the uncertainties associated with each of these quantities. When the drill bit reaches stage i , the estimates of bed boundary depths are updated. The stage-to-stage transition probabilities are modeled as a Markovian process with a constant correlation coefficient. Therefore, the estimation of bed boundary depth at stage $i+1$ is updated by determining the difference between the estimated and true depth of the bed boundary at stage i . With these updates, decisions can be made about steering the bit up or down to stay within zone. In the event of drilling out of zone, the team will need to decide whether the bit should be redirected into the reservoir by steering, or whether a sidetrack is a better option. These decisions are modeled by the DSDP to provide guidance towards the best-case scenario. To facilitate these decisions, the model is rewarded by keeping the wellbore within the reservoir and is penalized by drilling out of zone using a metric called the expected monetary value (EMV). The EMV of the project is determined by the length of the wellbore within the reservoir. Wellbores with more reservoir contact have a higher potential to access hydrocarbons, and therefore are assumed to be more valuable. In contrast, making the decision to sidetrack a well will incur additional costs not originally accounted for. However, in some situations, the decision to sidetrack a well to reach the target zone instead of re-steering may be more beneficial because of the possibility of having an additional length of wellbore in the target zone. Rapid decision making in these scenarios is challenging and [Kullawan et al. \(2017\)](#) demonstrate how the DSDP algorithm optimizes decision making in real-time.

In Esmersey et al. (2013), check shot measurements are required to look ahead of the bit, in addition to a reasonably accurate geologic model, neither of which are always available. In [Romanenkova et al. \(2019\)](#), the accuracy of the model is limited by the thickness of the layers and will have limited utility in highly heterogeneous formations with thin layers.

Mathematical Formulation

In this work, we use the concept of Hidden Markov Models (HMM) to predict changes in lithology from drilling data. This section describes how the hidden states (the lithology) can be detected from observations (surface drilling data). Note that we cannot directly observe the hidden states; instead, we hope to be able to infer them from surface drilling data. The fundamental concept underlying Hidden Markov Models is the Markov property which states that the current state of the system solely relies on the immediate past state and not the earlier history ([Gagniuc 2017](#)). This property is written as:

$$P(q_{t+1} | q_t, q_{t-1}, \dots, q_2, q_1) = P(q_{t+1} | q_t) \quad (1)$$

In other words, the next state $q_{(t+1)}$, is not dependent on the entire sequence history but only on the present q_t . A simple example of such states for a time-series of weather patterns is a sequence of sunny, cloudy and rainy days. Lithology variations can also be represented as a Markovian process because depositional processes typically tend to follow ordered patterns such as transgressive to regressive sequences. We now define a few key concepts that form the parameters of a HMM. We will discuss how to acquire these values later. First, for a transition from one state to another, we define a transition probability, written as:

$$a_{ij} = P(q_t = s_j | q_{t-1} = s_i) \quad (2)$$

where S_i and S_j refer to the current state and future state respectively. For a Markov chain with N states, there is an $N \times N$ matrix of transition probabilities. For the situation where there are only 3 states, this can be written as:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad (3)$$

where the sum of each row is equal to 1 (sum of transition probabilities from one specific state to all other possible states). Next, we also define an emission probability which is the probability of generating an observation when at state and this is given by:

$$b_{i(k)} = P(o_t = o_k | q_t = s_i) \quad (4)$$

For the case of three states and two observations, an emission probability matrix can be defined as:

$$B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix} \quad (5)$$

We also define an initial state probability of being in state, S_i at initial time and is given as:

$$\pi_i = P(q_1 = s_i) \quad (6)$$

If we only have three states, the initial probabilities are given by:

$$\pi = [\pi_1 \quad \pi_2 \quad \pi_3] \quad (7)$$

We now discuss the concept of a ‘hidden’ Markov chain. In a HMM, the process responsible for the states is hidden and not observed directly. However, we make observations generated by the underlying hidden process. Variations in lithology (hidden states) generating observations (surface drilling telemetry) is one such example. These observations (dependent on the states) are then used to identify a succession of states as shown in a directed graph of a first order Markov chain (Fig. 2). In our example, we wish to detect the sequence of lithologies or rocktypes using surface drilling data.

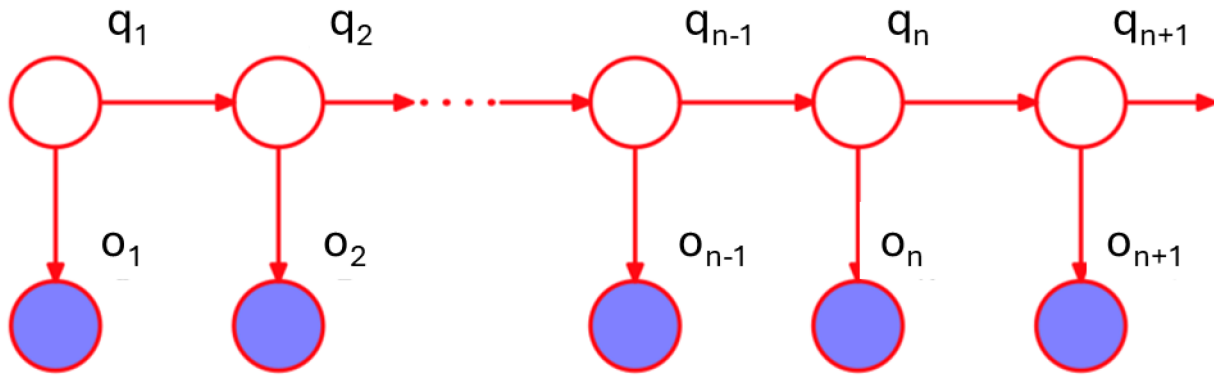


Figure 2—A sequential directed graph showing the basis of the HMM. Each state in the sequence represented by q is connected to the previous state, and an observation, o , occurring at the same time (modified from Bishop 2006).

In Fig. 2, the hidden, unobservable states represented by q , exhibit the first order Markov property of being connected to only the immediately preceding state. Each state generates an observation represented by o . The question now is whether we can predict the sequence of states, $\{q_1, q_2, \dots, q_N\}$, given the sequence of observations, $\{o_1, o_2, \dots, o_N\}$. In our specific problem, we are hoping to recover a sequence of lithologies or rocktypes given a time-series sequence of surface drilling data. While surface drilling data is a series of continuous observations, we show the mathematical formalism of the Hidden Markov Model using the example of discrete observations and states for ease of readability.

Following Rabiner and Juang (1986), we define a series of N possible states and a series of M possible observations.

$$\begin{aligned} S &= s_1, s_2, \dots, s_N \\ O &= o_1, o_2, \dots, o_M \end{aligned} \quad (8)$$

We also have the observation sequence over times, $1, 2, \dots, t-1, t, t+1, \dots, T$. With this notation, a HMM can be described as $\lambda = \langle \pi, A, B \rangle$, where λ represents the HMM.

At this time, we do not know any of the parameters of the HMM model. At any given time, we are only given a series of observations, $\{o_1, o_2, \dots, o_T\}$, using which we need to determine the three different parameters of the HMM model, which are π, A, B (the initial state probability, the state transition probabilities and the emission probabilities). We follow the work of [Petrushin \(2000\)](#) to provide a detailed overview of the HMM, beginning with the forward-backward algorithm. Note that in the following discussion, we are going to assume we know all the transition, emission and initial state probabilities. In a later section, we will discuss how to estimate those.

The Forward Algorithm

The forward algorithm is used to compute the probability that we are in state, $q_t = S_i$ at time, t given that we have observed a sequence of observations, $O_{(1:t)}$. We do not concern ourselves with how we got to state, S_i . Let us denote this probability as $\alpha_t(S_i)$, which is given by:

$$\alpha_t(S_i) = P(o_1, o_2, o_3, \dots, o_t, S_i | \lambda) \quad (9)$$

Therefore:

$$\begin{aligned} \alpha_t(S_i) &= \sum_{j=1}^N (\alpha_{t-1}(S_j) a_{ji}) b_i(o_t), \text{ for } 2 \leq t \leq T \text{ and } 1 \leq j \leq N \\ \alpha_1(i) &= \pi_i b_i(o_1), \text{ for } 1 \leq i \leq N \end{aligned} \quad (10)$$

To understand why [Eq. 10](#) holds, consider [Fig. 3](#), which shows a series of observations over two timesteps in the case where we have $N = 3$ and only 3 states, S_1, S_2 and S_3 . Starting at the initial time, the probability of generating observation o_1 , given we are in state, S_i is simply given by $\alpha_1(S_i)$ in the second term of [Eq. 10](#). For the next timestep, we can arrive at, for example, S_1 by considering three possible prior states and the corresponding transition probabilities.

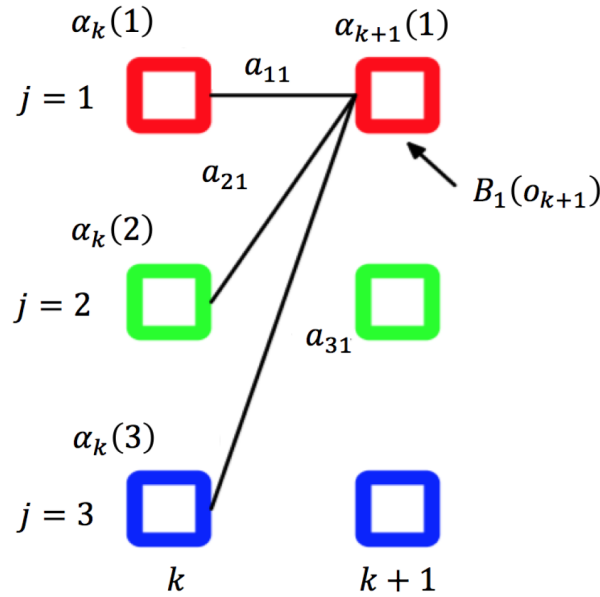


Figure 3—The forward recursion step of the forward backward algorithm for calculation of $\alpha_{(k+1)}(1)$. To calculate $\alpha_{(k+1)}(1)$, the values of $\alpha_k(j)$ are required which are calculated recursively starting with the initial probabilities π_j from the first time step forward ([Eq. 16](#)). Summing over the $\alpha_k(j)$ multiplied by their weights a_{ji} (transition probability from state j to 1), and multiplied by the emission probability B_1 for the observation occurring at time $k+1$ given state 1 allows for calculation of $\alpha_{(k+1)}(1)$.

The Backward Algorithm

The backward algorithm is used to compute the probability that we are in state, $q_t = S_i$ at time, t and generate the observations, $O_{(t+1:T)}$. Let us denote this probability as $\beta_t(S_i)$, which is given by:

$$\beta_t(S_i) = P(o_{t+1}, o_{t+2}, o_{t+3}, \dots, o_T | S_i, \lambda) \quad (11)$$

For the calculation of this variable, the calculations are done recursively backwards along the observation sequence. The backward calculation begins with an initialization step by setting $\beta_T(S_i) = 1$ for all states $i = 1$ to N . This is because we will observe o_T irrespective of the state we are in. Next, the recursion step is performed, which for the backward algorithm implies using $\beta_{(t+1)}$ to calculate β_t . This calculation is given by:

$$\beta_t(i) = \sum_{j=1}^N a_{ij} b_j(o_{t+1}) \beta_{t+1}(j), \text{ for } 1 \leq t \leq T-1 \text{ and } 1 \leq i \leq N \quad (12)$$

This calculation can be visualized similarly to the forward algorithm with a directed graph of states. Fig. 4 shows a graphical representation of the backwards algorithm to recursively calculate $\beta_k(i)$.

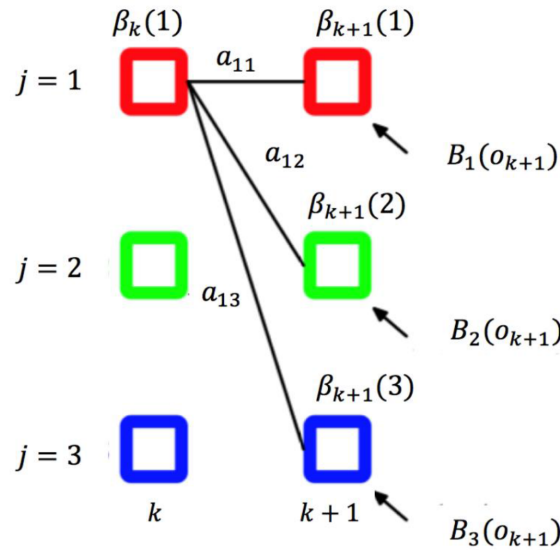


Figure 4—The backward recursion step of the forward backward algorithm for calculating $\beta_k(1)$. To calculate $\beta_k(1)$, the $\beta_{(k+1)}(i)$ values are required which are initialized as 1 for the final time step, and the subsequent values are recursively calculated backwards through the time steps. The emission probabilities $B_{(k+1)}(j)$ are required for each state, which are multiplied by their transition probabilities a_{ji} and summed then multiplied by $\beta_{(k+1)}(j)$.

Finally, we can compute the probability of the observing the sequence of observations $o_1, o_2, \dots, o_M$ as:

$$P(O|\lambda) = \sum_{i=1}^T \alpha_t(i) \beta_t(i) \quad (13)$$

Additionally, the probability of being in state, S_i at time t given the entire observation sequence $o_1, o_2, \dots, o_M$ is:

$$\gamma_t(i) = \frac{\alpha_t(i) \beta_t(i)}{P(O|\lambda)} \quad (14)$$

The Viterbi Algorithm

We now answer the question of the most likely sequence of states given a set of observations. This is a different question to the forward algorithm which merely asked what the likelihood is of being in a given

state having observed a series of observations. Again, we will use a recursive approach. If at any time, t , the most likely state is S_i , then the most likely state, S_j at time, $t+1$ is given by the probability:

$$\delta_{t+1}(S_j) = \text{MAX}_{S_i} [\delta_t(S_i) a_{ij}] b(o_{t+1} | j) \quad (15)$$

Starting with Eq. 16, we can then determine the most likely sequence of states.

$$\delta_1(S_i) = \pi_i b(o_1 | i) \quad (16)$$

We can now determine the most likely sequence of states given the observations.

The Baum-Welch Algorithm

Now that we have a formalism for computing the forward and backward probabilities as well as the most likely sequence of states, we can use the Baum-Welch algorithm (Baum, 1972) to determine the model parameters, π , A , B . We now answer the question of the most likely sequence of states given a set of observations. For the sake of brevity, we do not provide a full discussion of the approach. Instead, we refer the reader to the excellent article, Rabiner (1989).

Application to the Volve North Sea Field

In this section, we describe the use of Hidden Markov Models (HMM) on the Volve North Sea field data. The Volve field is an offshore decommissioned oilfield off the coast of Norway developed by Statoil and data related to this field is now publicly available (Equinor 2018). The field was discovered in 1993 by Statoil and commercially developed in 2008 and produced until 2016 (Equinor 2018). The field is estimated to have recovered 54% of the original oil in place at the point of abandonment, which was 3 to 5 years longer than the expected life of the field (Equinor 2018).

For each of the wells, we first estimate the electrofacies variations using K-means clustering (Gupta et al. 2017) that defines groups/clusters in multivariate data sets that have similar properties (MacQueen 1967). Fig. 5 shows the boxplots of log-derived properties for each electrofacies/rocktypes. Being able to demarcate variations in these electrofacies with depth using a HMM is the goal of this work.

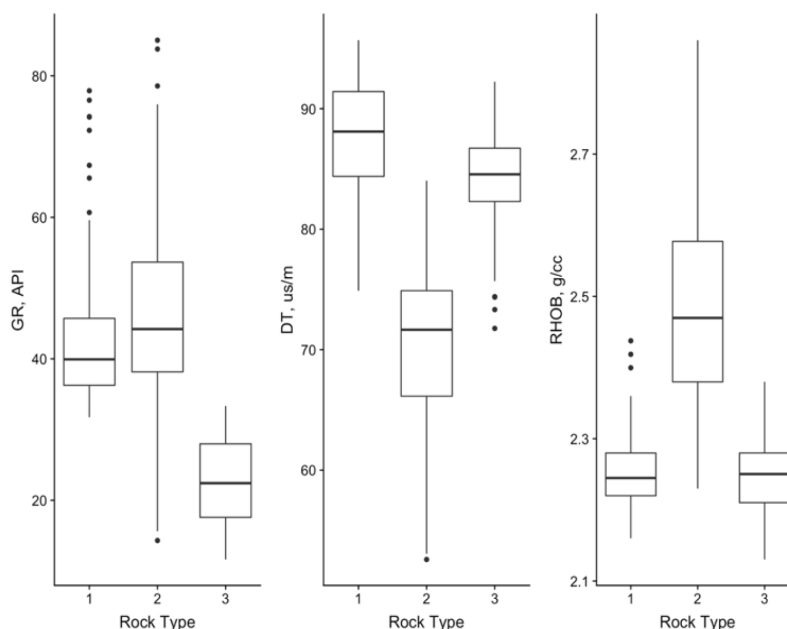


Figure 5—Boxplots showing the distribution of properties of the log-derived electrofacies/rocktypes derived using K-Means clustering. Each of the electrofacies/rocktypes has distinct log signatures.

We also apply a change-point detection (CPD) algorithm ([Aminikhanghahi and Cook 2017](#)) to the drilling data. Change point detection is a technique that attempts to identify changes within a random non-stationary signal. A random signal is considered to be second-order stationary if the mean and variance are constant over time ([Gagniuc 2017](#)). A non-stationary signal, on the contrary, is characterized by a spatially or temporally varying mean and variance. Examples of non-stationary signals include porosity as a function of depth in a multilayered reservoir with different facies, or hourly wind speeds recorded in weather surveys. [Fig. 6](#) is an example of CPD applied to time-series data showing transitions from one state to another. The use of CPD on surface drilling telemetry provides a visual guide to where time-series transitions are occurring, which in turn should, potentially, be linked to electrofacies transitions and transitions between HMM-derived states.

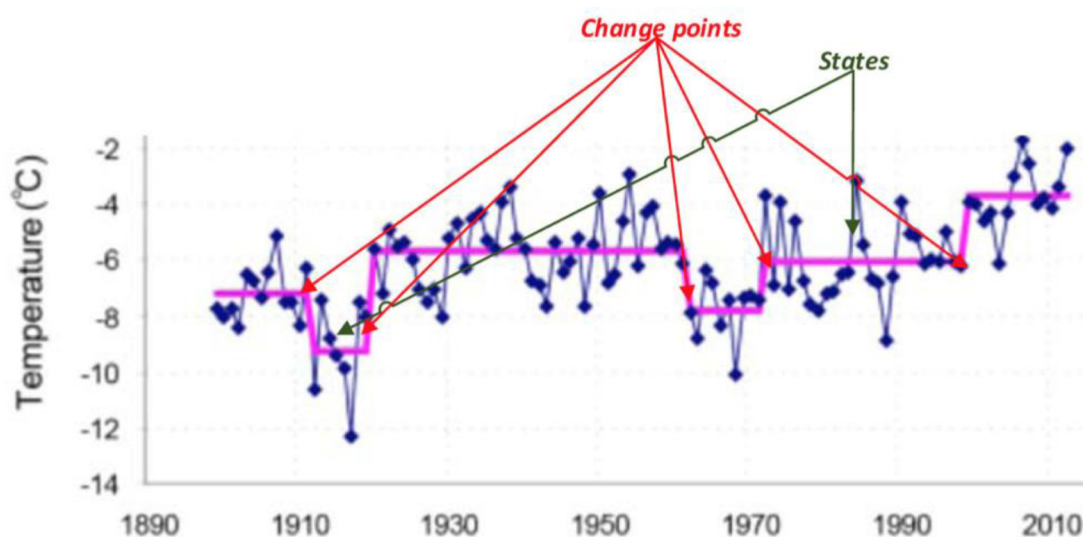


Figure 6—A plot showing a time series signal of yearly mean temperature of a Norwegian island. In this signal, the optimal segmentation τ is shown by different segments/states with the transitions between segments representing the change points. ([Aminikhanghahi and Cook 2017](#)).

An HMM model trained on the surface drilling telemetry provides the results shown in [Fig. 7](#) which plots the probability of each state at a specific depth. A higher probability for any specific state at a specific location indicates a higher likelihood of that specific state occurring. [Fig. 8](#) shows the drilling telemetry data colored by the highest probability HMM-derived state along with the predicted state transitions. There appears to be excellent correspondence between the occurrence of the change points and the derived HMM states. It is important to emphasize that both the HMM and change point detection algorithms are trained only on the drilling data and do not ‘see’ any electrofacies distributions. Consequently, our goal is to assess whether the change points and the HMM-derived states correspond to the electrofacies distributions.

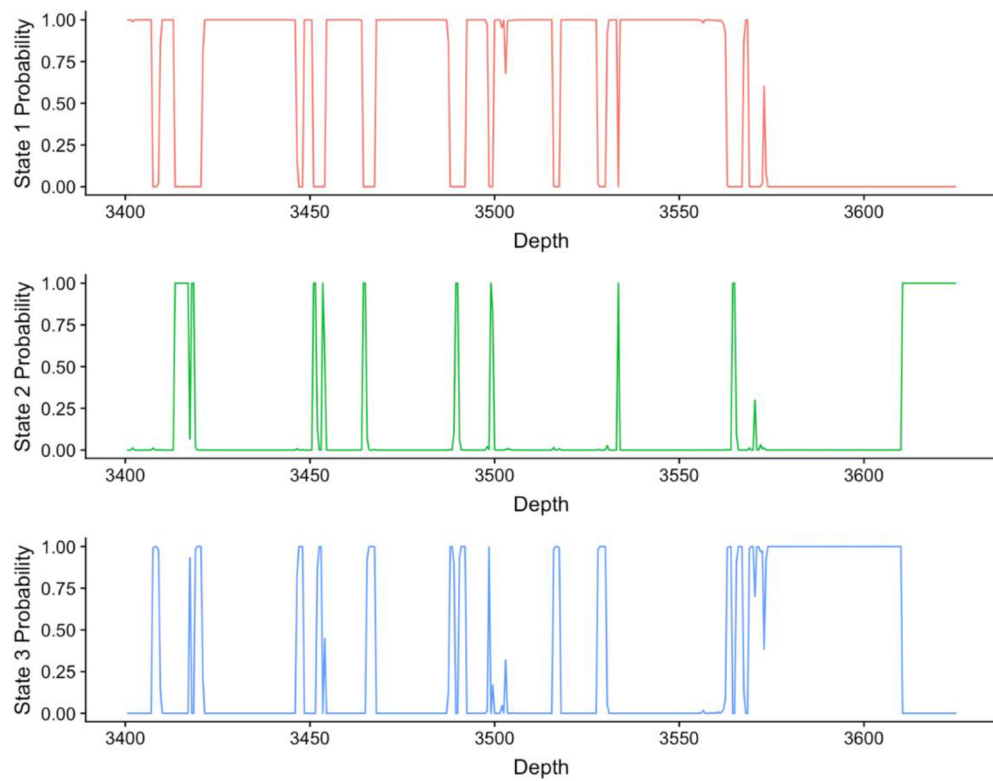


Figure 7—Plots showing the posterior probabilities of the HMM states from surface drilling data for well 15-4 plotted as a function of depth. The predicted HMM state is assumed to be highest probability state at each location.

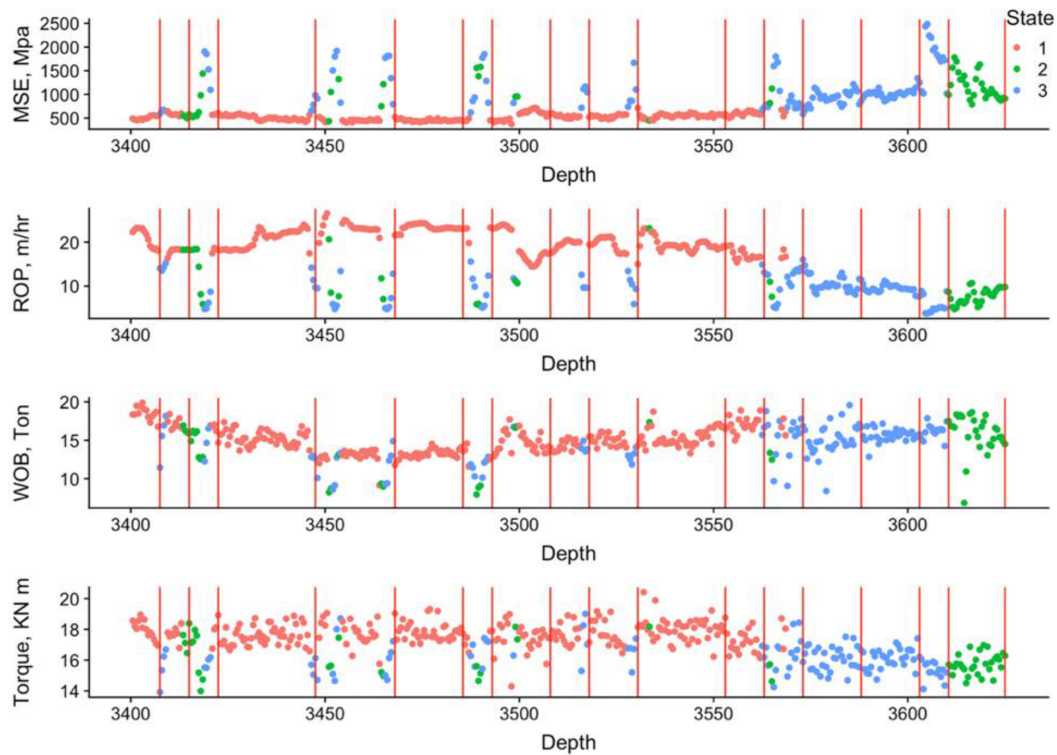


Figure 8—Plots showing the surface drilling data from well 15-4 in the Volve field. The data are colored by the HMM-derived states and independently derived change points shown with red vertical lines. There is a clear correspondence between where the change points occur and where the HMM-derived states transition, except between 3550m and 3600m, where more change points are detected while there are no HMM-derived state transitions.

With the excellent correspondence between where change points and HMM-derived state transitions occur, we need to assess whether these correspond to electrofacies transitions. In Fig. 9 we show gamma ray (GR) data colored by the HMM states on top and the same data colored by log-derived electrofacies below and also include change points. While there is a moderate degree of correspondence between the electrofacies variations and the trends in the HMM states, there are depths where the agreement is non-existent. While the weak agreement may be indicative of the inability of the HMM to sense electrofacies' variations in the drilling telemetry data, it is also important to indicate that the drilling telemetry itself is compromised.

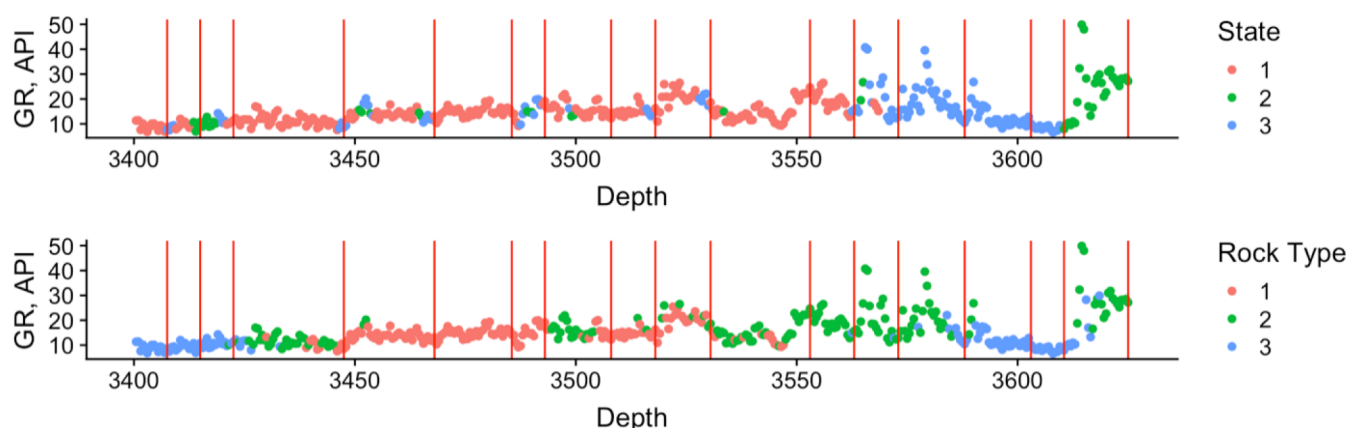


Figure 9—Plots showing the surface drilling HMM states, log-derived electrofacies, and change points for well 15-4. While the agreement between change points and state transitions was seen to be excellent in Fig. 8, there is a weak correspondence between the HMM-states and the electrofacies.

For instance, in Fig. 8, there are several state transitions between 3400m and 3550m that is not seen in the electrofacies/rocktype variations. Largely, these instances occur because of anomalous drilling data with spikes in the MSE, corresponding to sharp decreases in the ROP, torque and WOB, indicated by the green and blue colored states at those depths. These are instances when there appears to be some intentionality to the drilling operations over several tens of meters, ostensibly to avoid any drilling dysfunction. The original dataset (Equinor, 2018) does not provide a rationale for these short-lived changes in the drilling procedures. Nonetheless, because the HMM relies solely on drilling data, it becomes sensitive to these changes and identifies a change in state when no such transition occurs in the electrofacies' data.

Application to a Horizontal Unconventional Well

We next apply the HMM to a horizontal well in an unconventional play in Oklahoma in the US. The dataset comprises of surface drilling data for the lateral section of five wells as well as MWD-measured gamma ray values. Again, in the interests of brevity we only show the results for one well.

Because we do not have well log data, we can only compare HMM-derived states with the change points. However, Fig. 11 shows the GR log for the lateral section with the different states shown in different colors. The change points detected are shown as vertical red lines. Largely what we observe is the correspondence between the occurrence of the change points and the sharp increases or decreases in GR values, which are likely bed transitions. The HMM-derived data appears to show fewer transitions, and it becomes clear when viewing Fig. 10 again, we see that the HMM values largely correspond to different levels in the MSE data. While the HMM data does not closely match GR transitions, there is a strong correspondence with the MSE, which in turn is likely to be linked to rock mechanical properties and consequently, electrofacies variations.

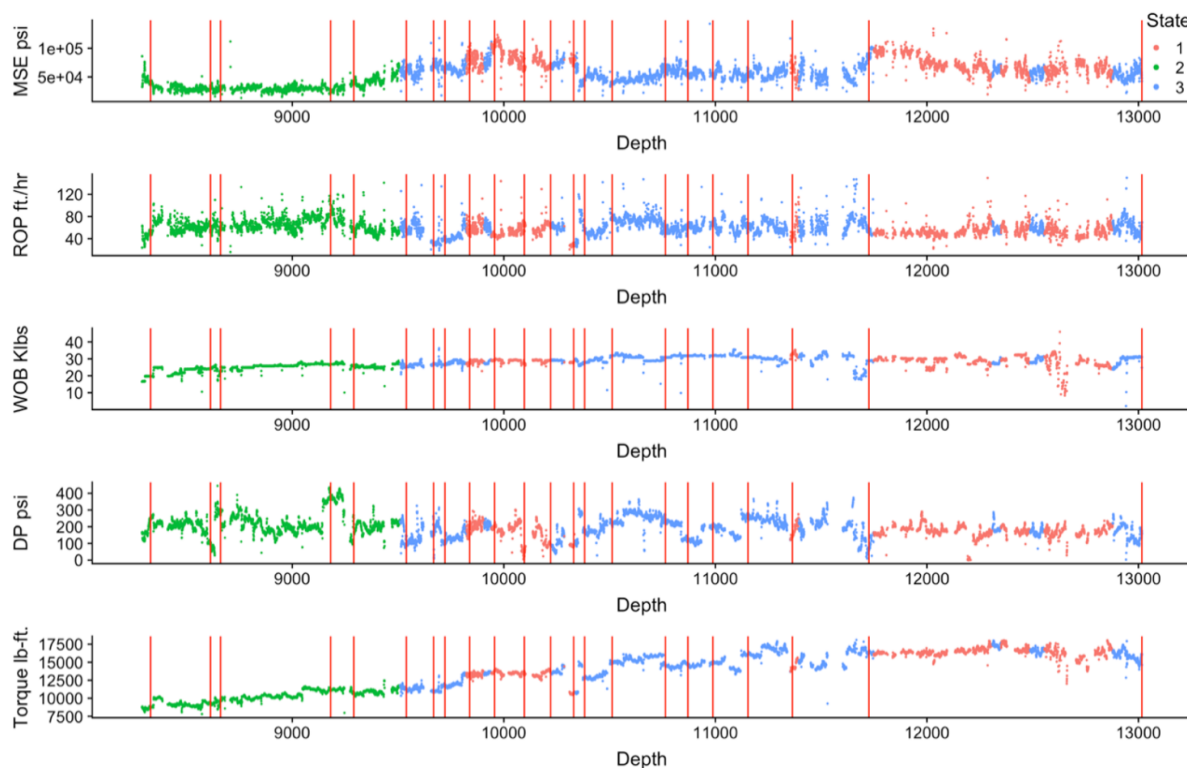


Figure 10—Plots showing the surface drilling HMM states and change points for the lateral section of a well in an unconventional play in Oklahoma. The change points appear to be excessively sensitive to small changes in the drilling telemetry.

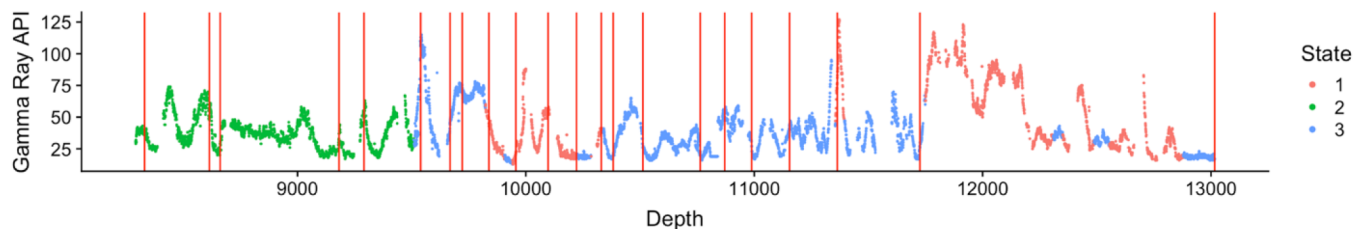


Figure 11—A plot showing the MWD measured gamma ray for the lateral section of a well in an unconventional play in Oklahoma colored by HMM state with the change points shown. The HMM states correspond to large transitions in the gamma ray values while the change points perform satisfactorily.

Predictive Capability of an HMM

An HMM can be used in a predictive fashion using the derived transition probabilities. Because this is of greater interest for vertical wells, we return to the Volve North Sea Field Study again. During drilling, an HMM can be fit to the available data up to the location of the bit. This includes estimation of the transition probabilities by the Forward-Backward and Baum-Welch algorithm as described earlier. Consequently, we can predictions on new real-time surface drilling data. This is illustrated in Fig. 12, where the HMM parameters are estimated using the drilling data available up to the black line. As and when data becomes available, we make a prediction on the next location. The figure on the top of Fig. 12 shows the actual states and the figure on the bottom shows the predictions given an HMM trained only a limited data (to the black line). Clearly there is excellent correspondence for an additional 75m, but beyond that we observe a weaker correspondence between the two charts.

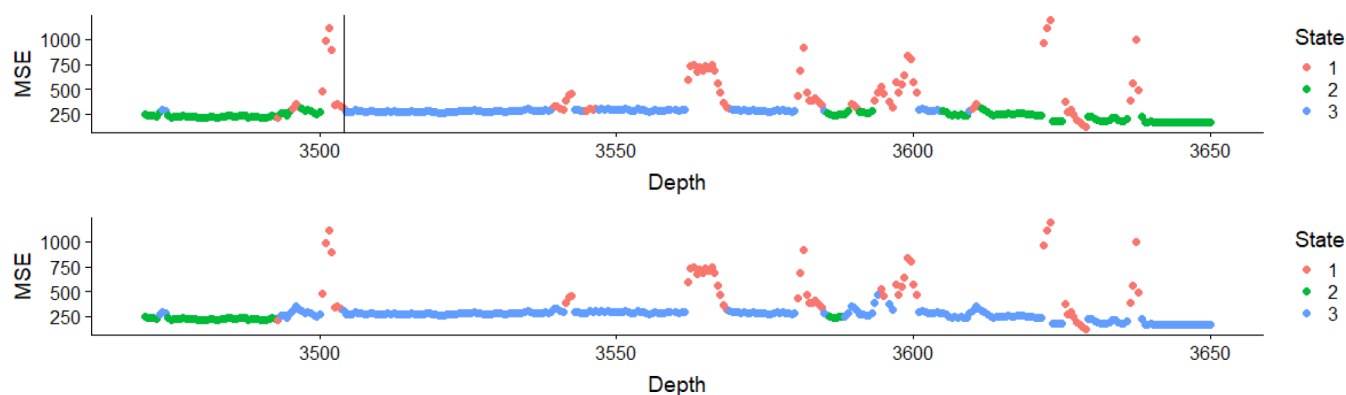


Figure 12—Predicted states for 150m ahead of the bit (black line) on the upper plot. The lower plot shows the true states. States closer to the black line (up to which training data was available) are predicted accurately, but farther away, the prediction accuracy drops. We see discrepancies between the predicted and true states beginning at 3575m.

This shows the excellent promise of a HMM in making predictions and identifying drilling states from streaming real-time drilling telemetry. In general, as shown in the examples above, there is a reasonable agreement between the HMM-derived states and the electrofacies/rocktypes underscoring the potential of HMM for real-time reservoir characterization using surface drilling telemetry.

Conclusions

In this work, we demonstrate the capabilities of state space modeling approaches, specifically the Hidden Markov Model (HMM) to identify state transitions in surface drilling data. The goal of the paper was to assess whether these states correspond to changes in rock properties as expressed by changes in electrofacies or rocktypes. The following conclusions can be drawn from this work:

1. HMM models trained on drilling telemetry provide reasonable accuracies in predicting state transitions that correspond to electrofacies/rocktype transitions.
2. We also show how change point detection can be utilized on multivariate time-series data to identify electrofacies/rocktype transitions. In general, change point detection can be sensitive to noise and can potentially show several false positives.
3. While the accuracy of the HMM is not perfect, it appeared to only fail when the drilling operation was intentionally controlled. These short-lived instances compromised the accuracy of the HMM and it stands to reason that the HMM will also be sensitive to changes in drilling operations, hole/bit size, drilling dysfunction and even type of bit or drilling company.

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